

Dharini Raghavan

Ph.D. Student, School of Electrical & Computer Engineering, Georgia Institute of Technology
draghavan7@gatech.edu · +1 (404) 729-7295 · Atlanta, GA · [Google Scholar](#) · [LinkedIn](#)

Research Profile

I develop representation-learning methods for biomedical imaging, generative modeling, and embodied perception. My research asks how images, signals, language, and geometry can be mapped into latent variables that are predictive, calibrated, mechanistically interpretable, and stable under distribution shift. The common thread across my work is state modeling: learning compact representations of biological or physical systems from partial, noisy observations.

Education

Georgia Institute of Technology <i>Ph.D. in Electrical & Computer Engineering; Advisor: Prof. Anant Madabhushi</i> Focus: medical imaging, generative modeling, clinical AI, computer vision.	May 2025–Present Atlanta, GA
Georgia Institute of Technology <i>M.S. in Electrical & Computer Engineering</i> Coursework: machine learning, MRI physics, biomedical sensing, advanced programming, MEMS.	Aug 2024–Dec 2025 Atlanta, GA
Ramaiah Institute of Technology <i>B.E. in Electronics & Communication Engineering</i> Graduated with Silver Medal; CGPA 9.76/10; Rank 2/156.	Aug 2019–May 2023 Bangalore, India

Research Experience

Madabhushi Lab, Georgia Tech & Emory University School of Medicine <i>Graduate Research Assistant; Advisor: Prof. Anant Madabhushi</i>	May 2025–Present Atlanta, GA
• LLM Reasoning in Medicine Investigating whether large language models perform diagnostic reasoning as structured inference or collapse toward prevalence priors under constrained clinical cases. The work frames medical LLM evaluation around constraint satisfaction, causal exclusion, evidence integration, and calibration under noisy observations. [ICLR'26 Workshop on Logical Reasoning of Large Language Models; Stanford AIMI Symposium] • Flow-based Generative Geometry Studying generative modeling through the geometry of probability transport. Analyzing how learned velocity fields, interpolants, and transport paths reshape distributions, how curvature and density concentration affect likelihood calibration, and how representation asymmetries emerge along the generative path. [NeurIPS'26 under review] • Physics-Consistent Vascular World Models Developing a vascular world model where the generative agent learns plausible, physics-consistent vascular states conditioned on tissue context, disease state, and intervention. The vascular morphology is treated as a structured generative object where branching topology, perfusion constraints, and treatment response are modeled as coupled state transitions rather than static image features. • Zero-shot Quality Representation Learning for Robust Vision Developing EyeQC, a zero-shot method for learning image-quality representations from frozen vision-language embeddings. The framework models acquisition artifacts as latent quality directions induced by semantic prompt contrasts, enabling continuous, multi-axis scoring of blur, illumination shift, contrast loss, occlusion, field-of-view restriction, and artifacts without supervised quality labels. In retinal imaging, this representation supports failure-mode decomposition, retrieval, batch-shift analysis, and quality-aware downstream learning. Extended the method to curriculum-based visual acuity prediction, using quality as both a confounder and a difficulty prior for robust regression. [ARVO'26 Hot Topic] • Oculomics for Systemic Risk Modeling Developing statistical-learning and representation-learning frameworks that use ocular structure as a non-invasive projection of systemic disease state. Current directions study disease-specific imaging biomarkers across retinal imaging modalities: OCTA capillary remodeling for coronary microvascular dysfunction [AJPC'26, ACC'26]; retinal vascular and perivascular geometry for cardio-renal-metabolic risk; deep fundus representations for pan cancer-risk modeling and cardio-vascular risk in cancer patients [ACC'26]; and 3D OCT-derived morphology for treatment-response modeling. • Frequency-Disentangled Generative Modeling for Topology-Preserving Synthesis Developed an unpaired cross-modal generation framework that factorizes image translation into low-frequency appearance transport and high-frequency anatomical topology preservation. The method constrains the generative map to preserve vessel geometry, connectivity, and microvascular structure while adapting modality-specific image statistics. Applied to fundus-to-OCTA synthesis, this reframes medical image generation as structure-preserving distribution alignment under weak supervision. • Spatial Risk-Field Learning for Surgical Margin Assessment Developing a multitask deep learning framework that treats surgical margin assessment in breast cancer resections as structured prediction over tissue geometry. The model learns coupled tumor-probability fields, margin-distance representations, morphology embeddings, and uncertainty maps, turning intermediate network states into interpretable measurements of residual disease risk. [Collaboration with	

ARPA-H]

ViTAL Lab, Georgia Tech & Emory University School of Medicine
Graduate Student Researcher; Advisor: Prof. Hyeokhyen Kwon

Aug 2024–Apr 2025
Atlanta, GA

- **Wearable AI for cognitive health** Developed multimodal sensing and interpretable machine-learning models for mild cognitive impairment using wearable IMU and ambient sensing. The work formulates gait, balance, and behavior as noisy longitudinal observations of cognitive and functional state, with emphasis on clinically interpretable digital biomarkers and subgroup robustness. [AMIA’25 Podium Presentation, PLOS Digital Health under review]
- **Robustness in clinical sensing** Analyzed how demographic heterogeneity, feature distributions, and clinical covariates shape prediction error in longitudinal sensing models, testing whether digital biomarkers remain stable across heterogeneous patient groups.

AI & Robotics Technology Park, Indian Institute of Science
ML Research Engineer; Advisor: Prof. Bharadwaj Amrutur

Apr 2024–Jul 2024
Bangalore, India

- **Zero-shot embodied localization** Developed robot localization methods combining foundation-model visual embeddings, keypoint geometry, and spatial consistency constraints, improving RTAB-Map SLAM accuracy by more than 63% without task-specific fine-tuning. [ACM ICVGIP’25]
- **Differentiable 3D scene representation** Advanced multi-camera pose refinement using anisotropic Gaussian primitives, SE(3) optimization, and differentiable rendering for infrastructure-assisted robot localization.

Autonomous Machines Lab, Indian Institute of Science
Research Assistant; Advisor: Prof. Bharadwaj Amrutur

Jul 2023–Mar 2024
Bangalore, India

- **Uncertainty-aware multi-robot perception** Designed fixed-camera localization systems for multiple robots under occlusion, clutter, and sensor noise, using ensemble disagreement, Monte Carlo dropout, and spatial uncertainty maps to quantify pose reliability.
- **Neural compression for edge perception** Applied structured pruning, INT8 quantization, and cross-architecture distillation for real-time 6-DoF pose estimation on Qualcomm RB5 platforms, achieving more than $3\times$ speedup with less than 2% accuracy degradation. [Qualcomm Developer Symposium]

NOKIA Networked Robotics Centre, Indian Institute of Science
Research Intern; Advisors: Prof. Bharadwaj Amrutur and Dr. Raghu Krishnapuram

Feb 2023–Jul 2023
Bangalore, India

- **High-precision visual localization** Developed infrastructure-camera localization for GPS-denied robot navigation using keypoint correspondences, geometric consistency, and 6-DoF pose estimation, enabling sub-centimeter positioning for safe human-robot collaboration.
- **Sim-to-real domain adaptation** Built synthetic-data generation pipelines with domain randomization over lighting, texture, pose, and occlusion, generating 100K+ annotated images and achieving less than 5% degradation on real-world deployment.

FIST Lab, Indian Institute of Science
Summer Research Fellow, Indian Academy of Sciences; Advisor: Dr. Jayaprakash

Jul 2021–Sep 2021
Bangalore, India

- **3D medical image deconvolution** Designed lightweight residual CNNs for non-blind 3D deconvolution of clinical MRI scans, improving PSNR by more than 2.5 dB over Richardson-Lucy baselines with more than $10\times$ faster inference. Extended the restoration framework to photoacoustic tomography for 3D tumor localization.

Indian Institute of Technology Indore
Research Intern; Advisor: Dr. Neelima Satyam

Sep 2021–Oct 2021
Indore, India

- **Geospatial ML for hazard prediction** Built ensemble-learning pipelines combining geospatial covariates, geological databases, and GeoTIFF imagery for landslide susceptibility mapping and spatial risk visualization.

Selected Projects

- **Generative state-space modeling of 4D cardiac dynamics** Exploring foundation-model methods for cine cardiac MRI that learn compact latent trajectories of cardiac motion, morphology, and remodeling from spatiotemporal image sequences. The project frames cardiac video as a generative state-space problem, combining representation learning, temporal attention, neural dynamics, geometry-aware embeddings, and counterfactual trajectory modeling to infer hidden physiological state.
- **Spatial pathology biomarkers in multiple myeloma** Investigating plasma-cell and immune-cell organization as a structured tissue system from bone marrow biopsies. The work uses deep learning and spatial tissue modeling to extract neighborhood geometry, local composition, immune-plasma interaction patterns, and architecture-level phenotypes associated with sex-specific survival differences. [AACR’26]
- **3D OCT biomarker detection** Developed multi-branch deep learning models for volumetric OCT biomarker classification, combining spatial/spectral feature streams, late fusion, and gradient-based interpretability; achieved $>98.5\%$

macro F1-score. [\[Code\]](#)

- **Wearable bioimpedance sensing for tendon health monitoring** Engineered real-time bioimpedance and IMU sensing for tendon strain and tissue-state discrimination. Modeled impedance variation as a physiologically constrained signal linked to tendon deformation and injury state.
- **Hybrid Bi-LSTM–Transformer for preterm birth prediction** Developed recurrent-attention models for spontaneous preterm-labor prediction from multi-channel electrohysterogram signals, combining temporal gating, self-attention, and neural compression; achieved >96% accuracy and >3× inference speedup. Preliminary benchmarking results published in [\[Biomedical Engineering: Applications, Basis and Communications’23\]](#)
- **Multimodal neuroimaging for Alzheimer’s disease** Developed machine-learning pipelines for Alzheimer’s disease classification using multimodal MRI–PET fusion and neuroimaging feature representations. [\[IEEE I4C’22; Journal of Machine Learning for Modeling and Computing’23\]](#)
- **Sparse ensemble networks for UAV detection and tracking** Designed sparse ensembles for miniaturized UAV detection, drone-bird discrimination, and RGB/IR localization under low illumination, clutter, and distortion; achieved >5× inference speedup and >25% relative localization improvement. [\[IEEE MysuruCon’22; IEEE ICASSP’23; IEEE ICIP Challenge’23; IEEE CONNECT’26 accepted\]](#)
- **NIR spectroscopy-based non-invasive glucometer** Designed a near-infrared sensing and neural calibration system for non-invasive glucose estimation using particle-swarm optimization and Clarke Error Grid validation. [\[TechRxiv’23\]](#)
- **Vision-guided robotic manipulation for sericulture** Implemented instance segmentation and 6-DoF inverse-kinematics control for autonomous sorting of diseased vs. healthy silkworms, integrating mask prediction, grasp localization, and class-conditioned manipulation. [\[IEEE ICDCECE’22; Patent published; Applied Research in Engineering Sciences’26\]](#)
- **Lightweight cardiac sound classification** Developed a lightweight deep ensemble model with attention mechanisms for cardiac sound classification, targeting efficient auscultation-based screening. Further converted this to an edge-to-cloud wearable system for vital-sign monitoring, PPG-derived HRV extraction, oscillometric blood-pressure estimation, and telemedicine integration. [\[Systems Science and Control Engineering’24\]](#)
- **Wearable textile antenna performance prediction** Developed ant-colony-optimized support vector regression models for performance prediction of wearable textile antennas. [\[IEEE GCITC’23\]](#)

Publications & Scholarly Presentations

Selected AI / ML Workshops, Challenges & Presentations

- **Dharini Raghavan.** “Deductive Constraint Satisfaction vs. Prevalence Priors: Benchmarking LLM Logic in Clinical Diagnostics.” *ICLR 2026 Workshop on Logical Reasoning of Large Language Models*; selected for Stanford AIMI Symposium presentation.
- **Dharini Raghavan.** “Target-adaptive representation routing for general CT diagnosis.” *CVPR 2026 Foundation Models for General CT Image Diagnosis Challenge*; ranked 3rd in validation phase.
- **Dharini Raghavan.** “Safe Active Medical Imaging: Theoretical Foundations for Uncertainty-Guided Acquisition under Safety Constraints.” *CVPR AVA Workshop 2025*, Nashville, TN. [\[Abstract\]](#)
- **Srijan Dokania, Dharini Raghavan.** “Zero-Splat TeleAssist: A Zero-Shot Pose Estimation Framework for Semantic Teleoperation.” *ICRA 3rd Workshop on Human-Centric Multilateral Teleoperation*, 2025. [\[Abstract\]](#)

Journal Article

- **Dharini Raghavan et al.** “Retinal OCTA-Derived Microvascular Remodeling Is Associated with Coronary Microvascular Dysfunction in Women with Ischemia and No Obstructive Coronary Artery Disease: A Pilot Study.” *American Journal of Preventive Cardiology*, 2026. [\[Paper\]](#)

AI + Clinical Meeting Abstracts

- **Dharini Raghavan et al.** “Quality-Aware Deep Learning Improves Prediction of Best Corrected Visual Acuity from Retinal Fundus Images.” *ARVO 2026*, Denver, CO. Highlighted as Hot Topic.
- **Dharini Raghavan et al.** “AI-Derived OCTA Retinal Biomarkers Detect Coronary Microvascular Dysfunction in Women.” *Journal of the American College of Cardiology*, ACC 2026, New Orleans, LA.
- **Dharini Raghavan et al.** “RETONCOMACE: An AI-Informed Cardio-Oncology Retinal Signature for Cardiovascular Risk Prediction in Cancer Patients.” *Journal of the American College of Cardiology*, ACC 2026, New Orleans, LA.
- **Dharini Raghavan et al.** “Sex-specific Plasma-Immune Architectural Differences in Bone Marrow Predicts Overall Survival in Multiple Myeloma.” *Cancer Research*, AACR 2026, San Diego, CA.
- **Dharini Raghavan et al.** “Quantifying Cognitive Decline and Balance in Older Adults with Mild Cognitive Impairment Using Wearables in Clinical Environments.” *AMIA 2025*, Atlanta, GA.

Peer-Reviewed Conference Papers

- **Dharini Raghavan**, Raghu Krishnapuram, Bharadwaj Amrutur. “Zero-Shot Pose Estimation and Tracking of Autonomous Mobile Robots using Infrastructure Vision Sensors: An End-to-End Perception Framework.” *ACM ICVGIP*, 2024. [Paper]
- **Dharini Raghavan** et al. “IR-SETNET: Sparsity Aware Ensemble Network for Infrared Imaging Based Drone Localization and Tracking in Distorted Surveillance Videos.” *IEEE ICIP Challenges and Workshops*, 2023. [Paper]
- **Dharini Raghavan** et al. “SETNET: A Sparse Ensemble Network for Drone Localization and Zero-Shot Drone Tracking in Real-Time Surveillance Videos.” *IEEE ICASSP Workshops*, 2023. [Paper]

Technical Expertise

Deep learning & foundation models: Representation learning, self-supervised and contrastive learning, vision-language models, LLM evaluation, diffusion models, flow matching, normalizing flows, GANs, curriculum learning, uncertainty estimation, model compression.

Biomedical AI & survival modeling: Medical foundation models, retinal imaging, OCT/OCTA, whole-slide pathology, CT, cine MRI, radiomics, spatial statistics, Cox models, Kaplan-Meier analysis, longitudinal modeling, bootstrap uncertainty, calibration.

Computer vision & embodied perception: Segmentation, detection, registration, image-to-image translation, pose estimation, SLAM, 3D Gaussian splatting, differentiable rendering, multi-camera perception, edge AI.

Programming & infrastructure: Python, PyTorch, TensorFlow/Keras, Hugging Face, scikit-learn, OpenCV, NumPy, pandas, SciPy, CUDA, Linux, Git, Docker/Apptainer

Sensing & hardware systems: IMU, PPG, bioimpedance, NIR spectroscopy, wearable sensing, embedded AI, Raspberry Pi, Arduino, Qualcomm RB5.

Honors & Awards

- **MIT Hacking Medicine Grand Hack 2025 – Winner, Assistive Technology Track;** Special Recognition: Google Health Prize and Niraj Bhatia Foundation Prize for Most Impactful Solution.
- **IEEE Signal Processing Society Travel Grants**, total \$2,000: PROGRESS Travel Grant for ICASSP 2023, Rhodes, Greece; SPS Travel Grant for ICASSP 2022, Singapore.
- **Best Project Award 2023**, Karnataka State Council for Science & Technology and Department of Science & Technology, Government of India.
- **Best Achiever Award**, Ramaiah Institute of Technology, Class of 2023.
- **Summer Research Fellowship 2021**, Indian Academy of Sciences, research at IISc.
- **MITACS Globalink Research Internship 2022**, selected for fully funded research internship at University of New Brunswick, Canada.
- **Avery Dennison Spirit of Invention Scholar 2021**, \$1,300 award for innovation with societal impact.
- **Winner, HMT SURGE Hackathon 2022**, IISc and Ministry of Heavy Industries, Government of India.
- **Winner, IEEE Eureka 2021**, Top 3 nationally among 600+ teams, IEEE Pune Section.
- **Academic Excellence:** Silver Medal; TATA Consulting Engineers Award.
- **Competition Placements:** 2nd Place, SIGPRO 2022, IEEE SPS; 2nd Place, Sovereign Nature Initiative Global Hackathon 2022.
- **Invitation to AI by HER Event**, Consulate General of India, Atlanta.

Patents

- **Dharini Raghavan** et al. “A System and Method for Autonomous Sorting of Healthy and Diseased Silkworm.” India Patent Application No. 202341026012, published 2023.

Professional Activities & Leadership

Invited Talks & Presentations

- Invited Lightning Talk, “Quality-Aware Deep Learning Improves Prediction of Best Corrected Visual Acuity from Retinal Fundus Images,” EyeConnect Symposium, Atlanta Vision Research Community.
- Research Presentation, Qualcomm Developer Symposium, neural compression for edge perception.
- Poster, “Neural Network Compression for Aerial Target Tracking,” IEEE ICASSP 2025, Hyderabad.

Professional Activities

- **Founder, Tech Talks:** launched and lead an AI journal club for critical discussion of recent papers in deep learning, computer vision, medical AI, foundation models, and generative modeling; organized reading sessions to translate frontier

methods into new research ideas and collaborations.

- Organizing Committee, VIP Cup: “Infrared-Visual Fusion for Drone Detection,” IEEE ICIP 2025, Anchorage, AK. [\[Website\]](#)
- Grand Challenge Organizer, “IR Imaging-Based Drone Tracking,” IEEE ICIP 2023, Kuala Lumpur. [\[Website\]](#)
- Student Volunteer, IEEE ICRA 2025, Atlanta, GA.

Peer Review

- NeurIPS’26, MICCAI’26, IEEE ICIP’26, IEEE ICIP’25, IEEE ICIP’24, IEEE ICASSP Workshops’23.
- *Informatics in Medicine Unlocked, Reliability Engineering & System Safety, Control and Cybernetics.*

Mentorship & Teaching

- Research Mentor: advising undergraduate researchers at Georgia Tech and Emory on machine learning for medical imaging projects.
- Hackathon Mentor & Judge, Smart India Hackathon 2023: mentored 40+ teams.
- K-12 STEM Outreach, IEEE SPS: hands-on robotics and signal-processing workshops for underrepresented students.

Leadership

- Secretary, IEEE MTT Society, 2022–2023.
- Mission Manager & Subsystem Lead, Student Satellite Team STARDUST RIT, 2020–2021.
- Executive Member, IEEE Student Branch, 2019–2021.
- Lead Editor, Annual Technical Newsletter, Department of ECE, Ramaiah Institute of Technology.
- Community Volunteer, COVID-19 Response, Caregiver Saathi Foundation.

Additional Information

Certifications: EFSET English Proficiency, C2 Level; Patent Search & Analysis, NPTEL; Microsoft Technology Associate.

Selected MOOCs: Machine Learning, Stanford/Coursera; Mathematics for Machine Learning: Linear Algebra, Imperial College London/Coursera; Python Programming, University of Michigan/Coursera.

Current mentees: Sanjana Shresta, Emory; Landon Minkow, Emory Mathematics; Viplav Patturi, Georgia Tech BME; Advay Balakrishnan, Georgia Tech CS.